

All we know about vector spaces

21 October 2023

Set theory

Sets

Set	A collection of things, called its <i>elements</i> (or <i>members</i>).
$x \in X$	“ x is an element of set X .”
$X \cup Y$	The union of X and Y : all elements of X together with all elements of Y .
$X \cap Y$	The intersection of X and Y ; its elements are those that are elements of both X and Y .
$X \setminus Y$	The set difference : all elements of X that are not elements of Y . Sometimes $X - Y$.
$X \subset Y$	“Every element of X is an element of Y .” X is a <i>subset</i> of Y .
$\{x \in X \mid P(x)\}$	The set of those elements of X satisfying the predicate P (i.e., for which $P(x)$ is true).
$\emptyset$	The empty set —the set with no members.

Pairs and tuples

Pair	An ordered list of two things: like a set but the order matters and they can be the same thing. For example: $(1, 2)$.
n-tuple	An ordered list of n things. For example: $(x_1, x_2, \dots, x_n)$.
$X \times Y$	The Cartesian product of X and Y is the set of all pairs (x, y) where $x \in X$ and $y \in Y$.
X^n	The set of all n -tuples of elements of X . The Cartesian product of X with itself n times.

Maps

Map	A rule which assigns, to every element of a set (called the <i>domain</i>), an element of another set (called the <i>codomain</i>). Sometimes called a <i>function</i> , especially when the codomain is numbers.
$f: X \rightarrow Y$	“ f is a map from X to Y .”
$f: x \mapsto y$	“Specifically, f maps the element $x \in X$ to the element $y \in Y$.”
$f \circ g$	The map g followed by the map f .
Injection	A map, $f: X \rightarrow Y$, is <i>injective</i> (also <i>one-to-one</i>) if no more than one element of X maps to a particular element of Y .
Surjection	A map, $f: X \rightarrow Y$, is <i>surjective</i> (also <i>onto</i>) if every element of Y is mapped to by some element of X .
Bijection	A map, $f: X \rightarrow Y$, is <i>bijective</i> if it is both injective and surjective (also “one-to-one and onto”).
$X \cong Y$	There exists a bijection between sets X and Y . Equivalently: X and Y are <i>isomorphic as sets</i> .

Useful maps

Operator	A <i>binary operator</i> on a set X is a map $\star: X \times X \rightarrow X$. That is, a binary operator takes two elements of X and returns an element of X . Operators are typically written in “infix” notation, $x \star y$, rather than in a function notation, $\star(x, y)$.
Associative	A binary operator, $\star$, is <i>associative</i> if

$$(a \star b) \star c = a \star (b \star c).$$

Almost all operators you will meet are associative. Since the order in which the operations are carried out doesn’t affect the result, the parentheses are often omitted, as in $a \star b \star c$.

Commutative	A binary operator, $\star$, is <i>commutative</i> if
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$$a \star b = b \star a.$$

Numbers

N	The <i>natural numbers</i> : $0, 1, 2, \dots$.
Z	The <i>integers</i> : $\dots, -2, -1, 0, 1, 2, \dots$.
Q	The <i>rationals</i> : All numbers that can be written as m/n where m and n are integers.
R	The <i>reals</i> : The rationals and “all the numbers in between.”
R⁺	The non-negative reals.
C	The <i>complex numbers</i> : Numbers of the form $a + bi$, where a and b are real numbers and $i^2 = -1$.

Vector spaces

Vector space A *real vector space* is a set, V , together with: (i) a commutative, associative, binary operator, $+$, on V ; (ii) a map, $\cdot: \mathbf{R} \times V \rightarrow V$; and (iii) a distinguished element $\mathbf{0} \in V$, such that:

1. $v + \mathbf{0} = v$ for all $v \in V$;
2. For any $v \in V$ there exists an element $-v \in V$ such that $v + (-v) = \mathbf{0}$;
3. $1 \cdot v = v$ for all $v \in V$;
4. $\alpha \cdot (\beta \cdot v) = (\alpha\beta) \cdot v$ for all $\alpha, \beta \in \mathbf{R}$ and all $v \in V$.
5. $\alpha \cdot (v + w) = \alpha \cdot v + \alpha \cdot w$ and $(\alpha + \beta) \cdot v = \alpha \cdot v + \beta \cdot v$ for all $\alpha, \beta \in \mathbf{R}$ and $v, w \in V$.

Replacing $\mathbf{R}$ with $\mathbf{C}$ in the above we obtain a *complex vector space*.

Vector An element of a vector space.

Subspace A subset of a vector space that is itself a vector space with respect to the addition and scalar multiplication inherited from the larger space.

Equivalently: A subset, $U \subset V$ of a vector space, V , is a *subspace* if, for all $u, v \in U$ and number α , the combination $u + \alpha \cdot v$ is also in U .

Examples of vector spaces

$\mathbf{R}^n$ (or $\mathbf{C}^n$) The set of n -tuples of $\mathbf{R}$ (or $\mathbf{C}$), together with the operation of “element-wise” addition:

$$(x_1, \dots, x_n) + (y_1, \dots, y_n) \stackrel{\text{def}}{=} (x_1 + y_1, \dots, x_n + y_n)$$

and multiplication by $\mathbf{R}$ (or $\mathbf{C}$):

$$\lambda \cdot (x_1, \dots, x_n) \stackrel{\text{def}}{=} (\lambda x_1, \dots, \lambda x_n)$$

$\mathbf{R}^X$ (or $\mathbf{C}^X$) For X a set, $\mathbf{R}^X$ denotes the set of all functions $X \rightarrow \mathbf{R}$, together with the operation of “pointwise addition”:

$$(f + g)(x) \stackrel{\text{def}}{=} f(x) + g(x)$$

and “pointwise multiplication by $\mathbf{R}$ ”:

$$(\alpha f)(x) \stackrel{\text{def}}{=} \alpha f(x).$$

(The notation “ $(f + g)(x)$ ” means “the function $f + g$, where $+$ is addition of functions, evaluated at the point x .”)

$\mathcal{P}_m(\mathbf{R})$ The set of polynomials of degree at most m . Thus, $f \in \mathcal{P}_m(\mathbf{R})$ implies $f(x) = a_0 + a_1x + \dots + a_mx^m$ for some m .

$\mathcal{P}(\mathbf{R})$ The set of all polynomials of finite degree.

Combining vector spaces

Sum (This definition is not commonplace.) For $U_1, U_2, \dots, U_n$ subspaces of V , their sum is the set of all sums of vectors from the U_i :

$$U_1 + \dots + U_n \stackrel{\text{def}}{=} \{v_1 + \dots + v_n \mid v_i \in U_i\}.$$

Equivalently, it is the set of all sums from $\cup_i U_i$ (since sums from within the same U_i are already elements of that U_i).

Equivalently, it is the span of $\cup_i U_i$.

Direct sum (Version 1: This definition is from Axler.)

The sum of subspaces, $U_1 + \dots + U_n$, is called a *direct sum* if the U_i satisfy the following property: if $v_1 + \dots + v_n = \mathbf{0}$, where $v_i \in U_i$, then each of the v_i is $\mathbf{0}$. If the sum of the U_i is a direct sum, it is written $U_1 \oplus \dots \oplus U_n$.

Direct sum (Version 2.) Suppose $U_1, \dots, U_n$ are vector spaces (not necessarily subspaces of some other space). Their *direct sum*, $U_1 \oplus \dots \oplus U_n$, is:

1. The set $U_1 \times \dots \times U_n$; together with
2. Addition and scalar multiplication given by

$$(u_1, \dots, u_n) + \alpha \cdot (v_1, \dots, v_n) = (u_1 + \alpha \cdot v_1, \dots, u_n + \alpha \cdot v_n).$$

Note that there are natural injective maps $U_i \rightarrow U_1 \oplus \dots \oplus U_n$ which “embed” the U_i as subspaces of the direct sum.

Linear independence and bases

Span	Let $v_1, \dots, v_n$ be vectors in a real vector space, V . The <i>span</i> of this set is the subspace given by $\{\alpha_1 v_1 + \dots + \alpha_n v_n \mid \alpha_1, \dots, \alpha_n \in \mathbb{R}\}$ (and likewise for a complex vector space).
Linear independence	Vectors $v_1, \dots, v_n \in V$ are <i>linearly independent</i> if $\alpha_1 v_1 + \dots + \alpha_n v_n = 0$ implies $\alpha_1 = \dots = \alpha_n = 0$.
Basis	(Of a vector space, V .) A collection of vectors that (a) spans V ; (b) is linearly independent.
Dimension	(Of a vector-space, V .) The number of elements of any basis of V . (Noting that any two bases of V have the same cardinality.)

Linear maps

Linear map	A map, $T: V \rightarrow W$ (where V and W are vector spaces) such that $T(u + v) = T(u) + T(v)$ and $T(\alpha u) = \alpha T(u)$ for all $u, v \in V$.
$\mathcal{L}(V, W)$	The set of all linear maps $V \rightarrow W$ with the vector space structure given by $(S + \alpha T)(v) \equiv S(v) + \alpha T(v)$ for any $S, T \in V$.
1_V	The identity map $1_V: V \rightarrow V$ where $1_V: v \mapsto v$.
$\circ$	Composition of linear maps. For linear maps $T: V \rightarrow W$ and $S: W \rightarrow X$, their composition $S \circ T$ is that linear map given by $(S \circ T)(v) \equiv S(T(v))$. Composition is associative and the identity map is a left and right identity.
Image	Of a linear map, that subspace of the codomain that is mapped to by <i>some</i> element of the domain. Sometimes called the <i>range</i> . However, range used to mean “codomain” so the term can be ambiguous.
Null space	(Or “kernel”.) Of a linear map, that subspace of the domain whose image is the subspace containing only the zero vector.
Inverse	For $T: V \rightarrow W$ a linear map, the <i>inverse</i> (if it exists) is the linear map T^{-1} such that $TT^{-1} = 1_V$ and $T^{-1}T = 1_W$.
Isomorphism	An isomorphism of vector spaces V and W is an invertible, linear map between the two. If an isomorphism exists the vector spaces are said to be <i>isomorphic</i> .

Quotient space

$a + U$	The set $\{a + u \mid u \in U\}$. What you get by “translating” U by a .
Coset	(Or “translate,” in Axler’s terminology.) For U a subspace of V , a coset of U in V is the set $a + U$ for some $a \in V$. Two cosets are either identical or disjoint. Every vector in V lies in one and only one coset.
Quotient	Of a vector space V by a subspace U : 1. The set of all cosets of U ; together with 2. The vector space structure given by $(a + U) + \alpha \cdot (b + U) = (a + \alpha \cdot b) + U$, noting that this formula does not depend on which representatives a and b are chosen.
V/U	The quotient of V by a subspace U .
Quotient map	The map, $\pi: V \rightarrow V/U$, which takes $v \in V$ to its coset $v + U$.

Dual space

Dual space The dual of a real (respectively, complex) vector space V is the vector space $V^* = \mathcal{L}(V, \mathbf{R})$ (respectively, $V^* = \mathcal{L}(V, \mathbf{C})$). An element of V^* is sometimes called a *linear functional*.

Dual basis For $(e_{(1)}, \dots, e_{(n)})$ a basis of V , the dual basis is that basis $(f^{(1)}, \dots, f^{(n)})$ of V^* such that

$$f^{(i)}(e_{(j)}) = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{otherwise.} \end{cases}$$

Dual map For $\phi: U \rightarrow V$ a linear map, its dual (sometimes called its transpose), is that map $\phi^*: V^* \rightarrow U^*$ such that, for all $u \in U$ and $\tilde{v} \in V^*$,

$$\tilde{v}(\phi(u)) = (\phi^*(\tilde{v}))(u).$$

Annihilator For $U \subset V$ a subset of V , the annihilator of U is that subspace $U^0 \subset V^*$ of the dual of V given by

$$U^0 = \{\tilde{u} \in V^* \mid \tilde{u}(w) = 0 \text{ for all } w \in U\}.$$

Operators I

Operator A linear map from a vector space to itself.

Invariant subspace A subspace $U \subset V$ is *invariant* under operator T if $Tu \in U$ for all $u \in U$.

Minimal polynomial Of an operator, T on a finite-dimensional vector space over field F . The (unique) monic polynomial $p \in \mathcal{P}(F)$ such that $p(T) = 0$. (“Monic” means that the coefficient of the highest-degree term is 1.)

Matrices

Matrix An $m \times n$ matrix A is $m \times n$ numbers, A_{ij} , indexed by $i \in \{1, \dots, m\}$ and $j \in \{1, \dots, n\}$. (Axler writes the elements as $A_{i,j}$.) Conventionally, the matrix is written as a rectangular array of the numbers, with A_{ij} being written in the i th row (starting at the top) and j th column (starting from the left). If $T: V \rightarrow W$ is a linear map and $\vec{e}_j \in V$ a basis of V and $\vec{f}_i \in W$ a basis of W , the **matrix of T** is given by:

$$T(\vec{e}_j) = \sum_i T_{ij} \vec{f}_i.$$

(That is, T_{ij} is the i th component of $T(\vec{e}_j)$ with respect to the basis $\vec{f}_i$.)

$\mathbf{R}^{m,n}$ The set of all $m \times n$ matrices over $\mathbf{R}$ (or $\mathbf{C}$, *mutatis mutandis*).

Addition of matrices and multiplication of matrices by numbers is defined elementwise:

$$(A + B)_{ij} \stackrel{\text{def}}{=} A_{ij} + B_{ij}$$

$$(\lambda A)_{ij} \stackrel{\text{def}}{=} \lambda A_{ij}.$$

Under these definitions, $\mathbf{R}^{m,n}$ is a vector space.

(The zero element of this space is the matrix

$$0_{ij} = 0.)$$

“Multiplication” of matrices is defined between an $l \times m$ and an $m \times n$ matrix, the result being an $l \times n$ matrix:

$$(AB)_{ij} = \sum_{k=1}^m A_{ik} B_{kj}.$$

Transpose The transpose of A_{ij} is the matrix A_{ji} .

Rank The rank of a matrix $A_{ij} \in \mathbf{R}^{m,n}$ is the dimension of the span of the n vectors $\vec{c}_j \in \mathbf{R}^m$ where $\vec{c}_j = A_{ij}$ (that is, the $\vec{c}_j$ are the “columns” of A_{ij}). The dimension of the span of the “rows” of A_{ij} is also the rank.

Upper triangular Of a matrix, having zero for all entries below the diagonal.

Diagonal Of a matrix, having zero for all entries except on the diagonal.

Eigenvalues

Eigenvalue Of an operator, T . A number λ such that there exists a vector $v \neq 0$ with $Tv = \lambda v$.

Eigenvector Of an operator, T . A non-zero vector, v , such that $Tv = \lambda v$ for some number λ .

Inner product spaces

Inner product On a real or complex vector space, V , a conjugate-symmetric, positive-definite map $V \times V \rightarrow \mathbb{F}$, written $\langle v, w \rangle$ for $v, w \in V$, which is linear in its first argument (and conjugate linear in its second). That is:

$$\langle v, v \rangle \geq 0 \quad (\text{positive})$$

$$\langle v, v \rangle = 0 \implies v = 0 \quad (\text{definite})$$

$$\langle v, w \rangle = \overline{\langle w, v \rangle} \quad (\text{conj. symm.})$$

$$\langle v, x + \lambda y \rangle = \langle v, x \rangle + \lambda \langle v, y \rangle \quad (\text{linear})$$

An **inner product space** is a vector space, V , together with an inner product on V .

Norm (Given an inner product) the norm of v is

$$\|v\| = \sqrt{\langle v, v \rangle}.$$

Most authors give an independent definition of a norm, from which it follows that the above construction gives rise to a norm, in those authors' sense.

The *Cauchy-Schwarz inequality* is that

$|\langle v, w \rangle| \leq \|v\| \|w\|$ with the inequality saturated only if v and w are colinear. The *triangle inequality* is that

$$\|v + w\| \leq \|v\| + \|w\|.$$

Orthogonal Describes two vectors, v and w , having the property that $\langle v, w \rangle = 0$.

Orthogonality

Orthonormal Of a list of vectors, having the property that every vector in the list has norm 1 and is orthogonal to every other vector in the list. Orthonormal lists are linearly independent.

Orthonormal basis A basis that is orthonormal.

Gram-Schmidt procedure The Gram-Schmidt procedure is an algorithm for constructing, from a list of vectors, an orthonormal list having the same span as the original list.

Orthogonal complement Of a subset $U \subset V$, the set, $U^\perp$, of all vectors in V that are orthogonal to every vector in U . Usually U is a subspace. When U is a finite-dimensional subspace, $(U^\perp)^\perp = U$.

Orthogonal projection Of a subspace U of V , the operator P_U define by $P_U v = u$ where $v = u + w$ and $u \in U$ and $w \in U^\perp$.

Pseudoinverse For V finite-dimensional and $T \in \mathcal{L}(V, W)$. The pseudoinverse $T^\dagger \in \mathcal{L}(W, V)$ is

$$T^\dagger w = (T|_{(\text{null } T)^\perp})^{-1} P_{\text{range } T} w.$$

Assorted results

Schur's theorem Every operator on a finite-dimensional, complex inner product space has an upper-triangular matrix with respect to some orthonormal basis.

Reisz representation theorem For finite-dimensional V , let $\phi \in V'$ be a linear functional on V . Then there exists a unique $v \in V$ such that

$$\phi(u) = \langle u, v \rangle$$

for every $u \in V$.

Operators II

Commuting Operators A and B commute if $AB - BA = 0$.
Adjoint